

1 Geodesics Are Locally Minimizing

We have known that every minimizing curve is a geodesic, now we turn to the converse situation.

Locally Minimizing Curves

Definition 1.1. : Let (M, g) be a Riemannian manifold, a regular curve $\gamma : I \rightarrow M$ is said to be **locally minimizing** if every $t_0 \in I$ has a neighborhood $I_0 \subseteq I$ such that whenever $a, b \in I_0$ with $a < b$, the restriction of γ to $[a, b]$ is minimizing.

Remark: It's easy to see that every minimizing admissible curve is locally minimizing. Our goal is to show that every geodesic is locally minimizing.

Recall: Recall that the differential map of the exponential map

$$d(\exp_p)_0 : T_0(T_p M) \cong T_p M \rightarrow T_p M$$

is precisely the identity map, i.e, $d(\exp_p)_0 = \text{Id}_{T_p M}$ is invertible. By the inverse function theorem for manifolds, we know that there exists a neighborhood V of $0 \in T_p M$ and a neighborhood U of $p \in M$ such that $\exp_p|_V : V \rightarrow U$ is a diffeomorphism. By this observation, the neighborhood U of p in M that is the diffeomorphism image under $\exp_p$ of a neighborhood of $0 \in T_p M$ is called a **normal neighborhood of p** .

Recall: **(Properties of Normal Coordinates)** Let (M, g) be a Riemannian n -manifold, let $(U; (x^i))$ be a normal coordinate chart centered at $p \in M$, then we have

(1) the coordinate of p is $(0, 0, \dots, 0)$;

(2) the component functions of the metric at p are $g_{ij} = \delta_{ij}$, i.e, $g = \delta_{ij} dx^i \otimes dx^j = \sum_{i=1}^n dx^i \otimes dx^i$ with respect to this normal coordinate;

(3) For any $v = v^i \partial_i|_p \in T_p M$, the geodesic γ_v starting at p with $(\gamma_v)'(0) = v$ is represented in normal coordinates by the line

$$\gamma_v(t) = (tv^1, tv^2, \dots, tv^n).$$

(Because of this property, the geodesics starting at $p \in M$ and lying in a normal neighborhood of p are called **radial geodesics**.)

(4) $\Gamma_{ij}^k(p) = 0$ in this normal coordinate, where Γ_{ij}^k is the Christoffel symbol.

If ε is a positive number such that $\exp_p$ is a diffeomorphism from the ball $B_\varepsilon(0) \subseteq T_p M$ to its image, then the image set $\exp_p(B_\varepsilon(0))$ is a normal neighborhood of p , called a **geodesic ball** in M , and $\exp_p(\partial B_\varepsilon(0))$ is called a **geodesic sphere**. In Riemannian normal coordinates centered at p , the geodesic balls and spheres are just coordinate balls and spheres.

Suppose U is a normal neighborhood of $p \in M$, given any normal coordinate (x^i) on U centered at p , define **radial distance function** $r : U \rightarrow \mathbb{R}$ by

$$r(x) := \sqrt{(x^1)^2 + \dots + (x^n)^2}$$

and define the **radial vector field** on $U \setminus \{p\}$, denoted by ∂_r , by

$$\partial_r := \frac{x^i}{r(x)} \frac{\partial}{\partial x^i}.$$

Recall: (Uniqueness of Normal Coordinate) Let (M, g) be a Riemannian n -manifold, $p \in M$ and U is a normal neighborhood of p . For every normal coordinate chart on U centered at p , the coordinate basis is orthgonal at p and for every orthonormal basis (b_i) for $T_p M$, there is a unique normal coordinate chart (x^i) on U such that $\partial_i|_p = b_i$ for $i = 1, 2, \dots, n$. Any two normal coordinate charts (x^i) and $(\tilde{x}^i)$ are related by

$$\tilde{x}^j = A_i^j x^i$$

for some matrix $(A_i^j) \in O(n)$.

Lemma 1.1. : *In everyn normal neighborhood U of $p \in M$, the radial distance function and the radial vector field are well defined, i.e., they are independent of the choice of the normal coordinates. Both r and ∂_r are smooth on $U \setminus \{p\}$ and r^2 is smooth on all of U .*

Proof: By the uniqueness of normal coordinates and computation we can show that r and ∂_r are independent of the choice of the normal coordinates. It's easy to see that r and ∂_r are smooth. $\square$

Gauss Lemma

Theorem 1.1. : *Let (M, g) be a Riemannain manifold, let U be a geodesic ball centered at $p \in M$ and let ∂_r denote the radial vector field on $U \setminus \{p\}$. Then ∂_r is a unit vector field orthogonal to the geodesic spheres in $U \setminus \{p\}$.*

Proof: For $\forall q \in U \setminus \{p\}$, let (x^i) be a normal coordinate on U , let $(q^1, q^2, \dots, q^n)$ denote the coordinate representation of q . Define $b := r(q) = \sqrt{(q^1)^2 + (q^2)^2 + \dots + (q^n)^2}$, where r is the radial distance function. Hence

$$\partial_r|_q = \frac{q^i}{b} \frac{\partial}{\partial x^i} \Big|_q.$$

Let $v := v^i \partial_i|_p \in T_p M$ with $v^i := q^i/b$. Thus the radial geodeisc with initial velocity v is given by $\gamma_v(t) = (tv^1, tv^2, \dots, tv^n)$ and $\gamma_v(0) = p$, $\gamma_v(b) = q$ and $\gamma'_v(b) = v^i \partial_i|_q = \partial_r|_q$. Since g_p is equal to the Euclidean metric in the normal coordinates, we have

$$|\gamma'_v(0)|_g = |v|_g = \sqrt{(v^1)^2 + \dots + (v^n)^2} = \frac{1}{b} \sqrt{(q^1)^2 + \dots + (q^n)^2} = 1,$$

which means that γ_v is a unit-speed geodesic, thus $\partial_r|_q = \gamma'_v(b)$ is also a unit vector.

Now we prove that ∂_r is orthgonal to the geodesic spheres. Let q, b, v be as above, $\Sigma_b := \exp_p(\partial B_b(0))$ be the geodesic sphere containing q . In these coorinates, Σ_b is the set of points satisfying the equation $(x^1)^2 + (x^2)^2 + \dots + (x^n)^2 = b^2$. Let $w \in T_q M$ be any vector tangent to Σ_b at q . We only need to show that $\langle w, \partial_r|_q \rangle = 0$.

Take a curve $\sigma : (-\varepsilon, \varepsilon) \rightarrow \Sigma_b$ such that $\sigma(0) = q$ and $\sigma'(0) = w$. Denote $\sigma(s) \triangleq (\sigma^1(s), \sigma^2(s), \dots, \sigma^n(s))$. Thus

$$(\sigma^1(s))^2 + \dots + (\sigma^n(s))^2 = b^2.$$

Now define a map $\Gamma : (-\varepsilon, \varepsilon) \times [0, b] \rightarrow U$ by

$$\Gamma(s, t) := \left(\frac{t}{b}\sigma^1(s), \dots, \frac{t}{b}\sigma^n(s)\right).$$

For each $s \in (-\varepsilon, \varepsilon)$, Γ_s is a geodesic with initial velocity $\Gamma'_s(0) = \frac{1}{b}\sigma^i(s)\partial_i|_p$, which is a unit vector. Thus Γ_s is a unit-speed geodesic.

Let $S := \partial_s \Gamma$ and $T := \partial_t \Gamma$. Hence we get

$$S(0, 0) = \frac{d}{ds} \Big|_{s=0} \Gamma_s(0) = 0$$

$$T(0, 0) = \frac{d}{dt} \Big|_{t=0} \gamma_v(t) = v$$

$$S(0, b) = \frac{d}{ds} \Big|_{s=0} \sigma(s) = w$$

$$T(0, b) = \frac{d}{dt} \Big|_{t=b} \gamma_v(t) = \gamma'_v(b) = \partial_r|_q$$

Thus $\langle S, T \rangle|_{(0,0)} = 0$ and $\langle S, T \rangle|_{(0,b)} = \langle w, \partial_r|_q \rangle$. we show that $\langle S, T \rangle$ is independent of t :

$$\frac{\partial}{\partial t} \langle S, T \rangle = \langle D_t S, T \rangle + \langle S, D_t T \rangle = \langle D_s T, T \rangle + \langle S, D_t T \rangle = \langle D_s T, T \rangle + 0 = \frac{1}{2} \frac{\partial}{\partial s} |T|^2 = 0$$

where the second equation is by the symmetry lemma and $|T| = |\Gamma'_s| = 1$ for all (s, t) . Hence we get $\langle w, \partial_r|_q \rangle = 0$, which proves the theorem. $\square$

Corollary 1.1. : *Let U be a geodesic ball centered at $p \in M$ and let r, ∂_r be the radial distance function and radial vector field defined as above. Then $\text{grad } r = \partial_r$ on $U \setminus \{p\}$.*

Proof: Recall that for any $f \in C^\infty(M)$, we have $\langle \text{grad } f|_p, w \rangle = df_p(w)$ for all $p \in M$ and all $w \in T_p M$. Recall that for $\forall f \in C^\infty(M)$ and $\forall X \in \mathfrak{X}(M)$ which is a nowhere-vanishing vector field on M , we have $X = \text{grad } f \iff Xf \equiv |X|_g^2$ and X is orthogonal to the level sets of f at all regular points of f . (see [1]P27 and P48.)

Apply these conclusions to r and ∂_r , by the Gauss Lemma, the corollary holds. $\square$

Proposition 1.1. : *Let (M, g) be a Riemannian manifold. $p \in M$ and q is contained in a geodesic ball around p . Then up to reparametrization, the radial geodesic from p to q is the unique minimizing curve in M from p to q .*

Proof: Let $q \in \exp_p(B_\varepsilon(0))$ and $\gamma : [0, c] \rightarrow M$ the radial geodesic from p to q parametrized by arc length, write $\gamma(t) = \exp_p(tv)$ for some unit vector $v \in T_p M$. Then we know $L_g(\gamma) = c$ since γ is of unit speed.

Let $\sigma : [0, b] \rightarrow M$ be another admissible curve from p to q , which is parametrized by arc length as well. Let $a_0 \in [0, b]$ denote the last time that $\sigma(t) = p$ and $b_0 \in [0, b]$ the first time after a_0 that $\sigma(t)$ meets the geodesic sphere Σ_c of radius c around p .

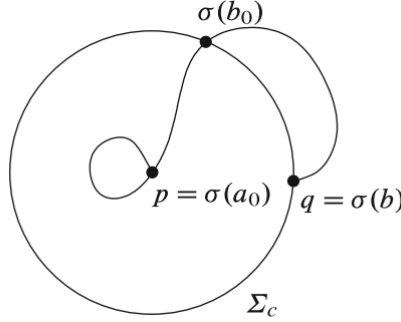


图 1: Radial geodesics are minimizing

By calculus, we get

$$\begin{aligned} r(\sigma(b_0)) - r(\sigma(a_0)) &= \int_{a_0}^{b_0} \frac{d}{dt} r(\sigma(t)) dt = \int_{a_0}^{b_0} dr(\sigma'(t)) dt = \int_{a_0}^{b_0} \langle \text{grad } r|_{\sigma(t)}, \sigma'(t) \rangle dt \\ &\leq \int_{a_0}^{b_0} |\text{grad } r|_{\sigma(t)} |\sigma'(t)| dt = \int_{a_0}^{b_0} |\sigma'(t)| dt = L_g(\sigma|_{[a_0, b_0]}) \leq L_g(\sigma), \end{aligned}$$

hence $L_g(\sigma) \geq r(\sigma(b_0)) - r(\sigma(a_0)) = c$, γ is minimizing.

Now suppose $L_g(\sigma) = c$, thus $b = c$ and both inequalities above are equalities, since we are assuming that σ is of unit speed, so we must have $a_0 = 0$ and $b_0 = b = c$. We can also know that

$$|\text{grad } r|_{\sigma(t)} |\sigma'(t)| - \langle \text{grad } r|_{\sigma(t)}, \sigma'(t) \rangle \equiv 0$$

on all of $[0, b]$, which implies that $\sigma'(t)$ is a positive multiple of $\text{grad } r|_{\sigma(t)}$ for all t . Since we are assuming σ has unit speed, so we must have $\sigma'(t) = \text{grad } r|_{\sigma(t)} = \partial_r|_{\sigma(t)}$. Thus σ and γ are both integral curves of ∂_r passing through q at time $t = c$, hence $\sigma = \gamma$. $\square$

The following corollaries show how the radial distance functions, balls, and spheres in normal coordinates are related to their global metric counterparts.

Corollary 1.2. : Let (M, g) be a connected Riemannian manifold, $p \in M$. Within every open or closed geodesic ball around p , the radial distance function $r(x)$ defined above is equal to the Riemannian distance from p to x in M .

Proof: Only needs to consider the open case. Let $x \in \exp_p(B_c(0))$, the radial geodesic from p to x is thus minimizing by the proposition before. Since $\gamma'(t) = \partial_r|_{\gamma(t)}$ is a unit vector in both g -norm

and the Euclidean norm in the normal coordinates, the g -length of γ is equal to its Euclidean length, i.e., $r(x)$. $\square$

Corollary 1.3. : *In a connected Riemannian manifold, every open or closed geodesic ball is also an open or closed metric ball of the same radius, and every geodesic sphere is a metric sphere of the same radius.*

Proof: For a Riemannian manifold (M, g) , let $p \in M$. Let $V \triangleq \exp_p(\overline{B}_c(0)) \subseteq M$ be a closed geodesic ball of radius $c > 0$ around p . For $\forall q \in M$, if $q \in V$, then the corollary above shows that q is also in the closed metric ball of radius c ; Conversely, suppose $q \notin V$, denote $S \triangleq \exp_p(\partial B_c(0))$ the geodesic sphere. The complement of S is hence the disjoint union of the open sets $\exp_p(B_c(0))$ and $M \setminus \exp_p(\overline{B}_c(0))$, which is disconnected. Thus for any admissible curve $\gamma : [a, b] \rightarrow M$ from p to q , there exists $t_0 \in [a, b]$ such that $\gamma(t_0) \in S$, by the corollary above we know that $L_g(\gamma|_{[a, t_0]}) \geq c$, which means that $d_g(p, q) > c$ since $L_g(\gamma|_{[t_0, b]}) > 0$, hence q is not in the closed metric ball of radius c around p .

Let $W \triangleq \exp_p(B_c(0))$ be an open geodesic ball of radius c . Since W is the union of all closed geodesic balls around p of radius c and the open metric ball of radius c is similarly the union of all closed metric balls of smaller radii, it's easy to know that W is equal to the open metric ball of radius c .

If $S \triangleq \exp_p(\partial B_c(0))$ is a geodesic sphere of radius c , by the argument above, we can see that S is equal to the closed metric ball of radius c minus the open metric ball of radius c , which is exactly the metric sphere of radius c . $\square$

Remark: By this corollary, we denote $B_c(p) := \exp_p(B_c(0))$, $\overline{B}_c(p) := \exp_p(\overline{B}_c(0))$ and $S_c(p) := \exp_p(\partial B_c(0))$ for open and closed geodesic balls and geodesic spheres, which we know are also open and closed metric balls and spheres.

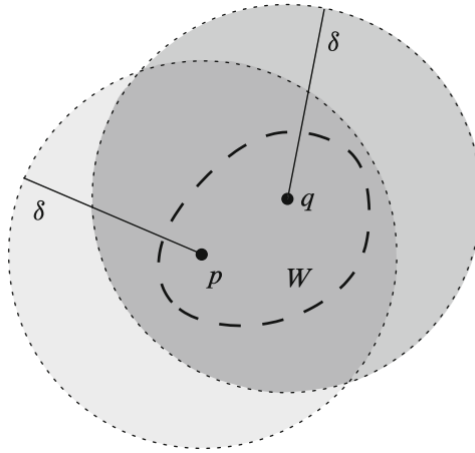


图 2: Uniformly normal neighborhood

Uniformly Normal Neighborhood

Definition 1.2. : Let (M, g) be a Riemannian manifold. A subset $W \subseteq M$ is called **uniformly normal** if there exists some $\delta > 0$ such that $W \subseteq B_\delta(q)$ for $\forall q \in W$, where $B_\delta(q)$ is the geodesic ball. We also say that W is **uniformly δ -normal**.

Uniformly Normal Neighborhood Lemma

Lemma 1.2. : Let (M, g) be a Riemannian manifold, $p \in M$ and U is an arbitrary neighborhood of p , then there exists a uniformly normal neighborhood of p contained in U .

Proof: Choose a normal coordinate chart $(U_0, (x^i))$ centered at p and contained in U and let (x^i, v^i) be the corresponding natural coordinates for $\pi^{-1}(U_0) \subseteq TM$. Since this is a local question, we identify U_0 with an open subset of $\mathbb{R}^n$ and identify TM with $U_0 \times \mathbb{R}^n$. Thus the exponential map for (U_0, g) is defined on an open subset $\varepsilon \subseteq U_0 \times \mathbb{R}^n$. Consider the map

$$E : \varepsilon \rightarrow U_0 \times U_0$$

$$(x, v) \mapsto E(x, v) := (x, \exp_x(v))$$

hence the differential of E at $(p, 0)$ is represented by the matrix

$$dE_{(p,0)} = \left(\begin{array}{cc} \frac{\partial x^i}{\partial x^j} & \frac{\partial x^i}{\partial v^j} \\ \frac{\partial \exp^i}{\partial x^j} & \frac{\partial \exp^i}{\partial v^j} \end{array} \right) \bigg|_{(p,0)} = \begin{pmatrix} \text{Id} & 0 \\ * & \text{Id} \end{pmatrix}$$

which is invertible, hence there exists neighborhoods $\mathcal{U} \subseteq U_0 \times \mathbb{R}^n$ of $(p, 0)$ and $\mathcal{V} \subseteq U_0 \times U_0$ of (p, p) such that $E|_{\mathcal{U}}$ is a diffeomorphism from $\mathcal{U}$ to $\mathcal{V}$. Shrinking both neighborhood if necessary, we may assume that $\mathcal{U}$ is a product set of the form $W \times B_\varepsilon(0)$, where W is a neighborhood of p and $B_\varepsilon(0)$ is a Euclidean ball in v -coordinates. It's clear that for each $x \in W$, $\exp_x : B_\varepsilon(0) \rightarrow \mathcal{V}_x := \{y \mid (x, y) \in \mathcal{V}\}$ is a diffeomorphism since we can know that its inverse is $\exp_x^{-1}(y) = \pi_2 \circ E^{-1}(x, y)$ where $\pi_2 : U_0 \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the projection.

We first prove a conclusion in the following yellow box, which will be used later in our proof.

Let g be a Riemannian metric on an open subset $W \subseteq \mathbb{R}^n$, and let $\bar{g}$ denote the Euclidean metric on W . For every compact subset $K \subseteq W$, there are positive constants c, C such that for all $x \in K$ and all $v \in T_x \mathbb{R}^n$, we have

$$c|v|_{\bar{g}} \leq |v|_g \leq C|v|_{\bar{g}}.$$

Consider the continuous function

$$F : TW \rightarrow \mathbb{R}$$

$$(x, v) \mapsto |v|_g = g_x(v, v)^{\frac{1}{2}}$$

for a compact subset $K \subseteq W$, define $L : \{(x, v) \in T\mathbb{R}^n \mid x \in K, |v|_{\bar{g}} = 1\}$, under the canonical identification $T\mathbb{R}^n \leftrightarrow \mathbb{R}^n \times \mathbb{R}^n$, L is equal to the product set $K \times \mathbb{S}^{n-1}$, which is clearly compact. Since F is positive on L , there are positive constants c, C such that $c \leq |v|_g \leq C$ for all $(x, v) \in L$.

For each $x \in K$ and $v \in T_x \mathbb{R}^n$ with $v \neq 0$, we write $v = \lambda \hat{v}$ where $\lambda = |v|_g$ and $\hat{v} = \frac{v}{\lambda}$, hence $(x, \hat{v}) \in L$ and thus we have

$$|v|_g = \lambda |\hat{v}|_g \leq \lambda C = C|v|_{\bar{g}}.$$

Argument in the other direction is similar;

Now, turn back to our proof, Shrinking W if necessary, we may assume that the metric g satisfies an estimate of the form above for all $x \in W$, which means that for each $x \in W$, the coordinate ball $B_\varepsilon(0) \subseteq T_x W$ contains a g_x -ball of radius at least ε/c . Take $\delta = \varepsilon/c$, so we have shown that for each $x \in W$, there is a g -geodesic ball of radius δ in M centered at x . Now, shrinking W once more, we may assume that its diameter w.r.t. the metric d_g is less than δ , it follows that for each $x \in W$ the entire set W is contained in the metric ball of radius δ around x . By Corollary 1.3 we know that this metric ball is also a geodesic ball of radius δ . Thus W is the desired uniformly normal neighborhood. $\square$

Theorem 1.2. : *Every Riemannian geodesic is locally minimizing.*

Proof: For a Riemannian manifold (M, g) , let $\gamma : I \rightarrow M$ be a geodesic, which we may assume to be defined on an open interval, let $t_0 \in I$. Suppose W is a uniformly normal neighborhood of $\gamma(t_0)$, let $I_0 \subseteq I$ be the connected component of $\gamma^{-1}(W)$ containing t_0 .

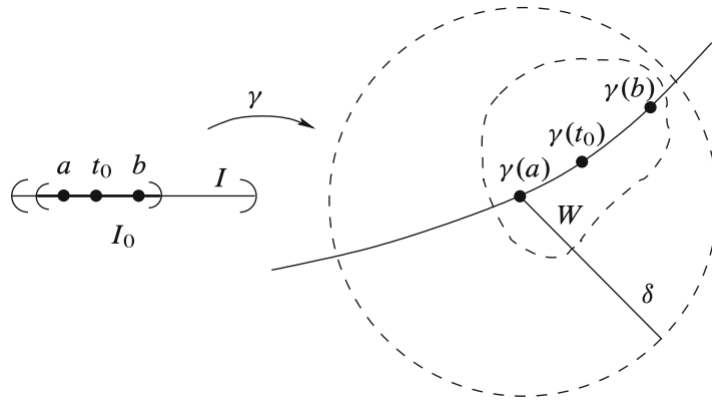


图 3: Geodesics are locally minimizing

If $a, b \in I_0$ with $a < b$, then the definition of uniformly normal neighborhood implies that $\gamma(b)$ is in a geodesic ball centered at $\gamma(a)$. By Proposition 1.1, the radial geodesic segment from $\gamma(a)$ to $\gamma(b)$ is the unique minimizing curve segment between these points. Since $\gamma|_{[a,b]}$ is also a geodesic segment from $\gamma(a)$ to $\gamma(b)$ lying in the same geodesic ball, hence $\gamma|_{[a,b]}$ coincides with this minimizing geodesic. $\square$

参考文献

- [1] John M. Lee *Introduction to Riemannian manifolds*
- [2] 陈维桓, 李兴校 黎曼几何引论 (上册)